Protocol Ports Reference Guide

Main Protocols from Wireshark Filters

HTTP (HyperText Transfer Protocol)

- **Port 80/TCP** - Standard HTTP port
- **Port 8080/TCP** - Alternative HTTP port (commonly used for web proxies, development servers)
- **Port 3128/TCP** - HTTP proxy servers
- **Port 8000/TCP** - Alternative HTTP port for development
- **Port 8443/TCP** - Alternative HTTPS port

HTTPS/TLS (Transport Layer Security)

- **Port 443/TCP** - Standard HTTPS port
- **Port 8443/TCP** - Alternative HTTPS port
- **Port 993/TCP** - IMAPS (IMAP over SSL/TLS)
- **Port 995/TCP** - POP3S (POP3 over SSL/TLS)
- **Port 465/TCP** - SMTPS (SMTP over SSL/TLS)
- **Port 636/TCP** - LDAPS (LDAP over SSL/TLS)

DNS (Domain Name System)

- **Port 53/UDP** - Standard DNS queries (primary)
- **Port 53/TCP** - DNS zone transfers and large responses
- **Port 5353/UDP** - mDNS (Multicast DNS) - used by Bonjour/Zeroconf

SSDP (Simple Service Discovery Protocol)

- **Port 1900/UDP** - SSDP multicast discovery
- **Multicast Address**: 239.255.255.250

TCP (Transmission Control Protocol)

TCP is a transport layer protocol that doesn't have specific ports itself, but provides the foundation for other protocols. The filters mentioned look at TCP flags (SYN, ACK) which are used regardless of the specific port.

Additional Common Protocols and Their Ports

Email Protocols

- **Port 25/TCP** - SMTP (Simple Mail Transfer Protocol)
- **Port 110/TCP** - POP3 (Post Office Protocol v3)
- **Port 143/TCP** - IMAP (Internet Message Access Protocol)
- **Port 587/TCP** - SMTP submission (authenticated SMTP)

File Transfer

- **Port 20/TCP** - FTP data transfer
- **Port 21/TCP** - FTP control
- **Port 22/TCP** - SFTP/SSH (Secure File Transfer Protocol)
- **Port 69/UDP** - TFTP (Trivial File Transfer Protocol)

Remote Access

- **Port 22/TCP** - SSH (Secure Shell)
- **Port 23/TCP** - Telnet
- **Port 3389/TCP** - RDP (Remote Desktop Protocol)
- **Port 5900/TCP** - VNC (Virtual Network Computing)

Network Services

- **Port 67/UDP** - DHCP server
- **Port 68/UDP** - DHCP client
- **Port 123/UDP** - NTP (Network Time Protocol)
- **Port 161/UDP** - SNMP (Simple Network Management Protocol)
- **Port 162/UDP** - SNMP traps
- **Port 514/UDP** - Syslog

Database Protocols

- **Port 1433/TCP** - Microsoft SQL Server
- **Port 1521/TCP** - Oracle Database
- **Port 3306/TCP** - MySQL/MariaDB
- **Port 5432/TCP** - PostgreSQL

- **Port 6379/TCP** - Redis
- **Port 27017/TCP** - MongoDB

Web Development & APIs

- **Port 3000/TCP** - Node.js development server (common default)
- **Port 4200/TCP** - Angular development server
- **Port 8080/TCP** - Tomcat, Jenkins
- **Port 9000/TCP** - Various development servers
- **Port 5000/TCP** - Flask development server (Python)

Streaming & Media

- **Port 554/TCP** - RTSP (Real Time Streaming Protocol)
- **Port 1935/TCP** - RTMP (Real Time Messaging Protocol)
- **Port 5004/UDP** - RTP (Real-time Transport Protocol)
- **Port 5005/UDP** - RTCP (RTP Control Protocol)

Gaming & P2P

- **Port 6881-6889/TCP** - BitTorrent
- **Port 4662/TCP** - eMule
- **Port 6346/TCP** - Gnutella

Port Ranges

Well-Known Ports (0-1023)

- Reserved for system services and require administrative privileges
- Examples: HTTP (80), HTTPS (443), SSH (22), DNS (53)

Registered Ports (1024-49151)

- Assigned by IANA to specific services
- Can be used by regular users
- Examples: SQL Server (1433), Oracle (1521)

Dynamic/Private Ports (49152-65535)

- Used for ephemeral connections

- Automatically assigned by the operating system for outgoing connections

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Wireshark Filter Examples with Ports

Filter by Specific Ports

```
tcp.port == 80      # HTTP traffic
tcp.port == 443     # HTTPS traffic
udp.port == 53      # DNS traffic
tcp.port == 22      # SSH traffic
```

Filter by Port Ranges

```
tcp.port >= 1024 and tcp.port <= 65535   # Non-privileged ports
tcp.port >= 49152                        # Dynamic ports
```

Combine Protocol and Port Filters

```
http and tcp.port == 8080          # HTTP on port 8080
dns and udp.port == 53             # Standard DNS queries
tls.handshake.type == 1 and tcp.port == 443  # HTTPS handshake
```

Notes for Network Analysis

1. **Default vs Alternative Ports**: Many services can run on alternative ports for security or configuration reasons.
2. **Ephemeral Ports**: Client connections typically use random high-numbered ports (ephemeral ports) to connect to server ports.
3. **Firewall Considerations**: Understanding these ports is crucial for firewall configuration and network security.
4. **Service Detection**: Port numbers help identify services, but port scanning can also be used maliciously.
5. **Protocol Independence**: Some applications can run on multiple ports or non-standard ports, so always verify the actual protocol being used.